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IDS 570

24 April 2026

A Trip to Paradise: Sentiment Analysis on Western Caribbean Ports

Introduction

Everyone needs a break at some point, and what better way to take that break than to board a cruise ship and sail away to paradise. The cruise industry is a popular and continuously evolving sector of travel and tourism. While cruise lines have successfully enticed guests with amazing restaurants, vibrant nightclubs, grand casinos, and immersive virtual reality experiences, most guests will not step foot onto a ship unless it is traveling somewhere interesting. Developing and monitoring an appealing itinerary is a crucial task for cruise lines, and it is one that requires immense consideration.

An ideal itinerary must include destinations that are visually stunning, safe, and filled with a variety of activities for guests, while remaining financially feasible for the cruise line. A common first step in determining destination viability is to review comments from travelers, including both cruise guests and those who have traveled by other means. These comments help cruise lines gain an initial understanding of traveler volume, guest demographics, and general sentiment, and seasonal trends.

Focusing on itinerary monitoring, this report explores the sentiment of Royal Caribbean guests toward specific ports in a Western Caribbean itinerary: Roatan, Honduras; Costa Maya, Mexico; and Cozumel, Mexico. By performing sentiment analysis on guest comments, this study

serves as a preliminary step to answering the core question: “How do Royal Caribbean guests feel about the ports in this Western Caribbean itinerary?” By evaluating these sentiments, Royal Caribbean can determine whether to retain these ports or seek new destinations to provide that once-in-a-lifetime experience for their guests.

Literature Review

The evaluation of guest satisfaction in the cruise industry increasingly relies on the automated processing of text data. A foundational component of this analysis is the selection of a robust sentiment engine. Coelho and Barbosa (2026) demonstrate the efficacy of the VADER (Valence Aware Dictionary and sEntiment Reasoner) framework for analyzing customer-oriented feedback, specifically in their study on patient complaints. While the primary domain is healthcare, the authors argue that VADER is uniquely suited for text found on social media because it accounts for both the intensity of sentiment and the nuances of colloquial language (Coelho & Barbosa, 2026). This study serves as a critical baseline, proving that VADER’s rule-based approach can successfully navigate the complexities of personal testimonials, which is a necessity when transitioning the technology to the maritime sector.

Building upon this, the application of VADER specifically to cruise guest reviews is validated through the lens of guest reactions to environmental attributes. Akter and Romanoff (2024) explore how specific physical and social environments on a ship influence internal emotional states. Their research provides the necessary industry context to bridge the gap between general feedback and cruise-specific sentiment (Akter & Romanoff, 2024). By applying VADER to cruise-centric data, this analysis confirms that the lexicon-based tool can be applied to explore cruise-based sentiment analysis, mirroring efforts to isolate which attributes drive guest satisfaction.

Technical challenges regarding the structure of tourism reviews necessitate advanced pre-processing, a topic addressed by Enache (2020) in her exploration of sentiment analysis within the tourism sector. Enache highlights that tourism sentiment is inherently comparative and often requires analysis at the sentence level (Enache, 2020). This project connects to Enache's findings by applying sentiment at the sentence and clause level rather than the document level. To address the challenge of comparisons, this analysis breaks sentences down into smaller clauses. This refinement improves both port assignment and the accuracy of sentiment values, ensuring that positive sentiment about a destination does not bleed into a separate, potentially negative comment about another destination.

The innovative spatial-temporal propagation of sentiment used in this analysis finds its theoretical counterpart in the work of Lai and Yan (2022) regarding asymmetric window multi-attentions. The authors introduce "asymmetric windows" to capture context that is not distributed equally around a target, suggesting that historical and future context should be weighted differently (Lai & Yan, 2022). The asymmetric window approach in this project, which utilizes specific ranges of sentences before and after a mention, directly operationalizes this theory. Just as they use multi-attentions to look back and look ahead, this analysis uses positional relationships to capture sentiment that appears slightly before a port is explicitly mentioned, as well as sentiment that follows it, acknowledging the narrative framing typical of guest storytelling.

Corpus

There are many sources Royal Caribbean can use to evaluate guest satisfaction, including internal surveys, social media posts, and public forums. However, due to proprietary restrictions and privacy concerns, this analysis uses a synthetic corpus of 600 reviews generated by Gemini 3

Flash and GPT-3. Two models were used to increase diversity in the responses. To aid with prompt generation, 15 responses from the *r/royalcaribbean* subreddit were selected as examples to provide insight on general review length, existing port activities, and genuine guest sentiments. No personal identifiable information was used in the process.

The prompt used to generate the synthetic corpus was highly structured and constraint-driven, designed to produce realistic yet controlled guest reviews centered specifically on port experiences. Reviews were generated in batches of 20 with strict output formatting requiring a valid JSON array. The prompt enforced a targeted distribution of review lengths, with 60% of responses between 100 and 200 words, 35% between 50 and 100 words, and 5% between 200 and 300 words. Additional constraints ensured that all reviews were written in English by adult guests on the same ship, and that each response referenced at least one target port. Onboard experiences were limited to brief contextual mentions to keep the focus on the destinations. Content controls further emphasized diversity in narrative perspective, sentiment, and traveler type. Since the reviews were processed line-by-line, the corpus was reformatted into a JSONL file. The reviews were generated between April 17 and April 20, 2026.

Methodology

The methodology is designed to attribute sentiment in cruise reviews to specific ports by leveraging the structured and predictable nature of the synthetic corpus, while considering robust alternatives. The pipeline begins by loading review data from a JSONL file using robust parsing logic that can handle malformed entries through fallback extraction when standard JSON parsing fails. Once loaded, the text is cleaned to ensure consistency, including normalization of punctuation variants, removal of encoding artifacts, and standardization of quotation and dash characters. Each review is then segmented into sentences using a spaCy model, and in some

cases further divided into clauses when contrastive structures such as “but” are detected. This clause level segmentation allows for finer-grained sentiment attribution within individual sentences, particularly when sentiment shifts occur. Ports are identified through regex matching against a predefined dictionary of canonical port names, and sentiment is computed at the sentence or clause level using VADER’s compound polarity score.

Once these core components are established, the key variation lies in how sentiment is propagated and assigned to ports based on detected mentions. In the Until Next Port approach, sentiment is assigned using a persistent context assumption that reflects the sequential structure of the dataset. When a port is mentioned, it becomes the active reference point, and all subsequent sentences or clauses are attributed to that port until a new port is introduced. This approach leverages the structured nature of the corpus, where reviews typically focus on one port at a time before transitioning to the next. By maintaining this running context, sentiment can be accumulated across multiple sentences without requiring repeated port mentions, aligning with the consistent narrative flow of the data.

The Forward Window approach introduces a more constrained mechanism by limiting sentiment attribution to a fixed local range following each port mention. Each detected port activates a window that includes the current sentence and the next two sentences, and sentiment is only assigned within this range. This design is based on the assumption that the most relevant sentiment about a port is typically expressed shortly after it is introduced. To reduce noise from weak or less informative sentiment signals, a minimum sentiment threshold is applied so that only stronger polarity values contribute to attribution. This results in a more localized assignment strategy that prioritizes precision by restricting the temporal spread of sentiment around each port mentioned.

The Asymmetric Window approach assigns sentiment using a bidirectional context window around each detected port mentioned. For every occurrence of a port, sentiment is attributed to sentences within a range extending one sentence before the mention and two sentences after it. Port positions are first indexed across the review, and these indices are then used to determine which surrounding sentences fall within the attribution window. This allows the method to capture sentiment that may appear slightly before a port is explicitly mentioned, as well as sentiment that follows it, reflecting the contextual nature of narrative framing in travel reviews. Methods such as clause splitting and minimum sentiment thresholding are not included in this approach in order to reduce unintended noise and improve the interpretability and consistency of results across port assignments.

Results and Discussion

Analysis of Average Sentiment

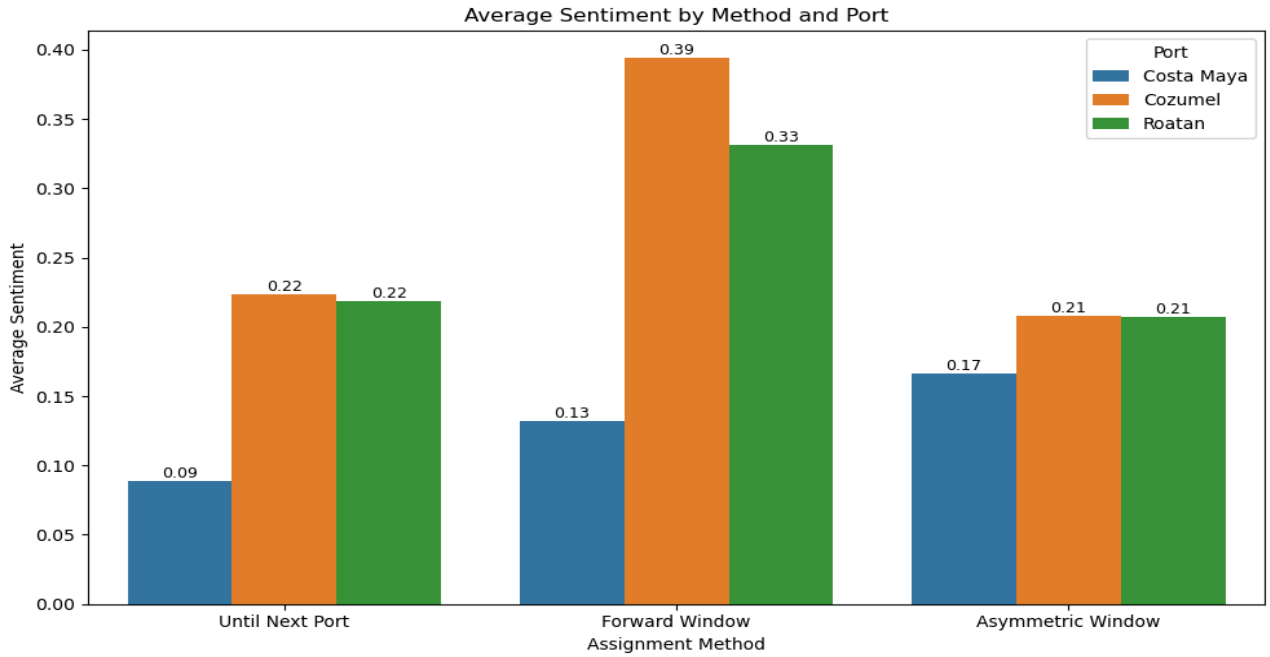

Figure 1. Comparative Mean Sentiment by Port and Methodology

The initial analysis of the sentiment data reveals that all three ports on the itinerary maintain a positive average sentiment score. This suggests that, broadly speaking, cruise guests are satisfied with their experiences at these destinations. However, a granular look at the data reveals significant disparities between the locations. Costa Maya consistently exhibits lower sentiment scores than Cozumel and Roatan across all three assignment methodologies.

The consistency of this trend suggests that the lower performance of Costa Maya is likely a reflection of guest perception. These findings align with qualitative reviews observed on the *r/royalcaribbean* subreddit. A primary complaint regarding Costa Maya is the lack of engaging activities in the immediate vicinity of the port. While the region is famous for Mayan Ruins, these excursions often require a commute of over an hour each way. For many guests, the prospect of spending a significant portion of their limited time on a bus, coupled with excursion costs ranging from $80 to $130 per person, results in a perceived lack of value. Guests who are not interested in Mayan history or who are unaware of alternatives often find themselves staying on the ship or confined to the immediate port area, leading to the depressed sentiment scores visualized in Figure 1.

Methodological Comparisons

review_id	port	sentence	sentiment
1	Roatan	Roatan ended up being the highlight for me.	0.34
1	Roatan	I booked a small group tour that included a wildlife sanctuary and a short beach stop.	-0.296
1	Roatan	The sloth interaction felt a little touristy	0
1	Roatan	still memorable, and the guides were genuinely funny and engaging.	0.6486
1	Roatan	The beach portion was short, maybe too short,	0
1	Roatan	the water was calm and clear.	0.5994
1	Costa Maya	In Costa Maya, we tried to explore on foot	0
1	Costa Maya	quickly realized everything near the port is built for cruise passengers.	0
1	Costa Maya	We ended up taking a taxi to a quieter beach, which was better	0.4404
1	Costa Maya	felt rushed because of limited port time.	-0.2263
1	Cozumel	Cozumel was the easiest to navigate-we just walked around town, grabbed lunch, and did some light shopping.	0.4215
1	Cozumel	Overall, I liked the variety of stops	0.296
1	Costa Maya	wished Costa Maya had either more time or better access to less commercial areas.	0.4877

Table 1: Port Assignment and Sentiment from the Until Next Port Approach

review_id	port	sentence	sentiment
1	Roatan	Roatan ended up being the highlight for me.	0.34
1	Roatan	I booked a small group tour that included a wildlife sanctuary and a short beach stop.	-0.296
1	Roatan	still memorable, and the guides were genuinely funny and engaging.	0.6486
1	Cozumel	Cozumel was the easiest to navigate-we just walked around town, grabbed lunch, and did some light shopping.	0.4215
1	Cozumel	Overall, I liked the variety of stops	0.296
1	Costa Maya	wished Costa Maya had either more time or better access to less commercial areas.	0.4877

Table 2: Port Assignment and Sentiment from the Forward Window Approach

review_id	port	sentence	sentiment
1	Roatan	Roatan ended up being the highlight for me.	0.34
1	Roatan	I booked a small group tour that included a wildlife sanctuary and a short beach stop.	-0.296
1	Costa Maya	The sloth interaction felt a little touristy but still memorable, and the guides were genuinely funny and engaging.	0.7876
1	Costa Maya	The beach portion was short, maybe too short, but the water was calm and clear.	0.7469
1	Costa Maya	In Costa Maya, we tried to explore on foot but quickly realized everything near the port is built for cruise passengers.	0
1	Cozumel	In Costa Maya, we tried to explore on foot but quickly realized everything near the port is built for cruise passengers.	0
1	Costa Maya	We ended up taking a taxi to a quieter beach, which was better but felt rushed because of limited port time.	-0.1027
1	Cozumel	We ended up taking a taxi to a quieter beach, which was better but felt rushed because of limited port time.	-0.1027
1	Costa Maya	We ended up taking a taxi to a quieter beach, which was better but felt rushed because of limited port time.	-0.1027
1	Cozumel	Cozumel was the easiest to navigate-we just walked around town, grabbed lunch, and did some light shopping.	0.4215
1	Costa Maya	Cozumel was the easiest to navigate-we just walked around town, grabbed lunch, and did some light shopping.	0.4215
1	Cozumel	Overall, I liked the variety of stops but wished Costa Maya had either more time or better access to less commercial areas.	0.7046
1	Costa Maya	Overall, I liked the variety of stops but wished Costa Maya had either more time or better access to less commercial areas.	0.7046

Table 3: Port Assignment and Sentiment from the Asymmetric Window Approach

When comparing the Until Next Port and Asymmetric Window approaches in Figure 1, Cozumel and Roatan show remarkably similar sentiment levels. This similarity is somewhat surprising given the differences in how these methods handle text. The Until Next Port method, aided by clause-splitting, appears to provide the most accurate representation of the synthetic corpus because it accurately maps most of the sentences to the correct port (shown in Table 1).

Conversely, the Asymmetric Window approach struggled with the synthetic data (shown in Table 3). This weakness likely stems from window overlap between multiple ports, which can cause the same sentence to be assigned to multiple locations. Furthermore, the lack of clause-splitting and the retention of "noisy" or weak sentiment sentences likely diluted its accuracy. Despite these struggles on the synthetic corpus, the Asymmetric Window approach may actually be more robust for real-world reviews, which lack the rigid structure of

AI-generated text. Additionally, it is possible that in the current results, Cozumel and Roatan misattributed sentiment to one another, causing their averages to balance out near the Until Next Port benchmarks.

A notable shift occurs with Costa Maya in the Asymmetric Window approach, where positive sentiment nearly doubles compared to its sentiment in the other methods. This is likely a result of misattribution, where positive sentiment intended for neighboring ports was incorrectly assigned to Costa Maya due to the windowing constraints.

The Forward Window approach displays the highest degree of variation among the averages, likely due to its ability to drop sentences that are considered weak in sentiment. While this effort to refine the data might suggest superior performance, it carries a significant risk of data loss. By restricting the sentiment capture to only two succeeding sentences after a port is mentioned, the method likely ignores relevant sentiment that falls outside this narrow boundary. Additionally, since the analysis relies on the VADER dictionary, terms that a human might recognize as sentiment-bearing but which VADER considers neutral are discarded. The Forward Window approach serves as a middle ground. While Table 2 indicates it performs better at assignment than the Asymmetric Window, it captures less total sentiment than the Until Next Port approach. The latter retains all sentences, ensuring no sentiment is lost, but at the cost of retaining unrelated noise.

Port Assignment Proportions

The distribution of assignments for each port remains relatively stable across the methodologies, as seen in Figure 2.

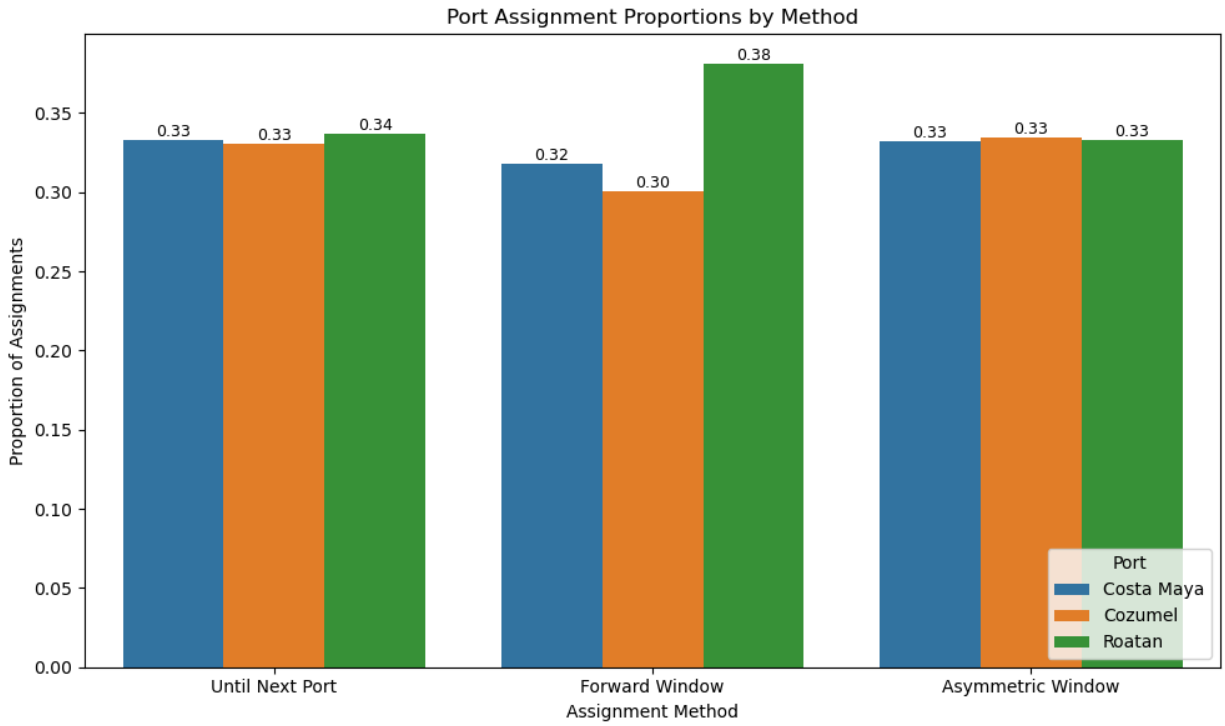

Figure 2. Distribution of Port Assignment for each Assignment Method

The Until Next Port and Asymmetric Window approaches show nearly identical distributions, while the Forward Window approach shows slight deviation. The uniformity observed is likely a byproduct of the synthetic nature of the corpus. Because the LLMs were given consistent prompts and word count constraints, the overall distribution of sentiment remained even. This explains why even when sentiment was misattributed, the resulting proportions remained stable. The variation in the Forward Window approach is specifically tied to its mechanism of dropping weak sentiment sentences, which filtered the data differently than its counterparts.

Recommendations

To improve guest satisfaction and optimize the itinerary, the following actions are recommended for Royal Caribbean:

- **Implement Aspect-Level Sentiment Analysis:** Royal Caribbean should prioritize the implementation of aspect based sentiment analysis as the most critical next step for itinerary optimization. While the current project operates at the port level, research by Enache and Akter and Romanoff suggests that moving toward this more granular approach is essential for generating deeper, actionable insights. By integrating this methodology, the cruise line can move beyond generalities to identify the specific features of a destination (such as transportation, safety, pricing, or excursion variety) that are driving underperformance in locations like Costa Maya.

  Isolating these specific attributes will allow stakeholders to make data driven decisions on whether a port should remain on an itinerary with targeted, feasible adjustments or be replaced entirely. For instance, determining if guest dissatisfaction stems primarily from the commute time to the ruins versus the actual cost of the activity provides a clear roadmap for operational changes. Ultimately, addressing these linguistic and structural complexities will refine the ability to transform raw guest feedback into precise intelligence for cruise line deployment.

- **Diversify and Advertise Local Excursions:** Royal Caribbean should increase marketing efforts for non-Mayan Ruin excursions that are closer to the port. Highlighting local culture or beach activities that do not require a two-hour bus ride may appeal to guests who currently find the port uninteresting.
- **Expand Near-Port Offerings:** The cruise line should actively seek out or develop new excursion partnerships within a shorter radius of the port to provide better value for the guests' time and money.

- **Guest Experience Surveys:** A targeted survey should be distributed to guests to identify popular private excursions. Understanding what guests are finding on their own can help Royal Caribbean curate better official offerings that align with guest preferences.

Limitations

Several limitations must be considered when interpreting these results:

Synthetic Data Constraints

The use of a synthetic corpus created a highly structured and predictable environment that may not mirror the linguistic messiness of real guest reviews. Although precautions were taken, such as opening new LLM conversations every two batches and using multiple models, the sheer volume of 300 responses per model makes some level of repetition inevitable. This predictability likely smoothed out some of the nuances that would be present in authentic human feedback.

Asymmetric Approach Implementation

Unlike the other two methods, the Asymmetric Window approach did not receive special adjustments like clause-splitting. Implementing these features created conflicts with the existing constraints of the windowing logic, making it difficult to track how sentences were being assigned. Had these adjustments been successfully integrated to look at previous sentences without clashing with the window boundaries, the alignment accuracy likely would have improved.

Sentiment Lexicon Limitations

The study relied on the standard VADER dictionary, which is not tuned for the specific nuances of the cruise industry. For instance, the term "touristy" often carries a negative connotation in travel reviews, yet VADER categorizes it as a neutral term. The absence of a

custom, industry-specific sentiment library means that certain qualitative nuances were likely missed or categorized as neutral, potentially skewing the final sentiment scores.

Furthermore, as noted in the literature, the reliance on rule-based lexicons like VADER introduces inherent difficulties in detecting irony and sarcasm. Addressing these linguistic complexities remains a significant hurdle in transforming raw text into actionable intelligence for cruise line deployment.

Conclusion

This analysis demonstrates that while the Western Caribbean remains a generally well-regarded itinerary, guest satisfaction is not uniform across all destinations. By utilizing VADER-based sentiment analysis and various windowing methodologies, this study addressed the question of how Royal Caribbean guests perceive their time in Roatan, Cozumel, and Costa Maya. The findings reveal a clear preference for the vibrant activities of Cozumel and Roatan, while highlighting a significant sentiment deficit in Costa Maya, largely driven by the perceived lack of accessible, high-value excursions. While the methodologies varied in precision, the consistent underperformance of Costa Maya across all models provides a clear signal for operational review. Ultimately, for a cruise line to maintain its reputation for excellence, it must ensure that every stop on the map meets the high expectations of its guests, because in the competitive world of travel, paradise is a standard that must be met at every single port.

References

Akter, Sabina, and Jani Romanoff. “Cruise Passengers’ Internal Reactions to Onboard Environmental Attributes.” *Ocean Engineering*, vol. 307, 2024, article 118112, doi:10.1016/j.oceaneng.2024.118112.

Coelho, João Vasco, and Liliana da Costa Barbosa. “Enhancing Public Healthcare Through VADER Sentiment Analysis: A Case Study on Patient Complaints.” *Journal of Patient Experience*, vol. 13, 2026, doi:10.1177/23743735251413861.

Enache, Maria. “Sentiment Analysis in Tourism.” *Annals of Dunarea de Jos University of Galati. Fascicle I: Economics and Applied Informatics*, vol. 26, 2020, pp. 81–85, doi:10.35219/eai1584040999.

Lai, H., and X. Yan. “Multimodal Sentiment Analysis with Asymmetric Window Multi-Attentions.” *Multimedia Tools and Applications*, vol. 81, 2022, pp. 19415–19428, doi:10.1007/s11042-021-11234-y.